

Spreading Out Across Expanding Idea Space

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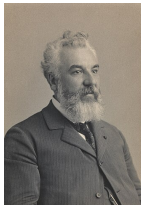
University of Essex

October 2026

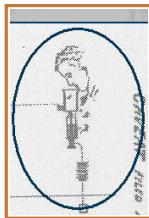
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February 14, 1876: A Collision in Idea Space



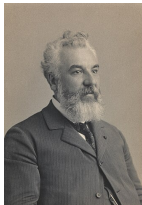
Alexander Graham Bell



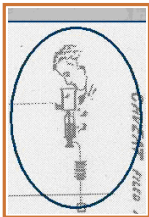
Elisha Gray

A **patent interference** — two inventors, same invention

February 14, 1876: A Collision in Idea Space



Alexander Graham Bell



Elisha Gray

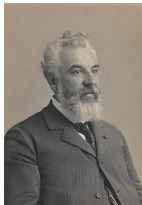


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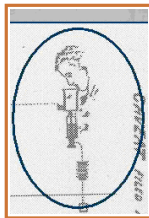


If every invention a
"location" in idea space

February 14, 1876: A Collision in Idea Space



Alexander Graham Bell



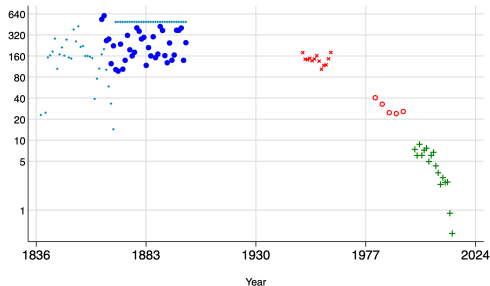
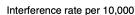
Elisha Gray



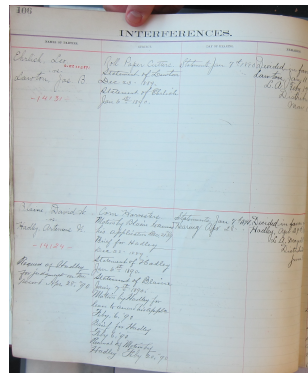
A **patent interference** — two inventors, **same invention** $\longleftrightarrow$ They chose the **same location** in idea space

Such Collisions Were Once Routine

≈ 1 in 20 patents (1870s) → <1 in 10,000 (2014)



• Finding Guide • Registers • ✖ Di Simone et al. • ○ Calvert-Sofocleous • + eFOIA decisions

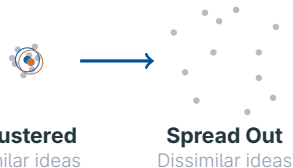
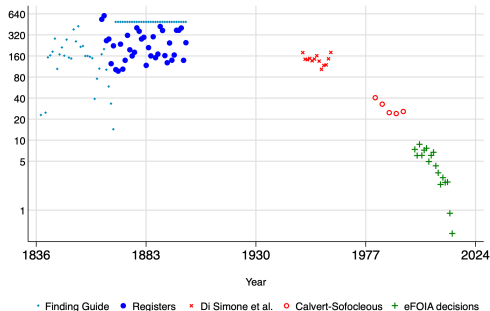


- ▶ Data

Such Collisions Were Once Routine

≈ 1 in 20 patents (1870s) $\rightarrow$ <1 in 10,000 (2014)

Interference rate per 10,000



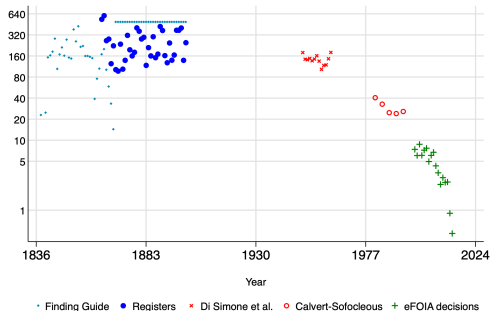
Inventors spread out
across expanding idea space

► Data

Such Collisions Were Once Routine

≈ 1 in 20 patents (1870s) $\rightarrow$ <1 in 10,000 (2014)

Interference rate per 10,000



Clustered
Similar ideas



Spread Out
Dissimilar ideas

Inventors spread out
across expanding idea space

Their positions
matter for growth

► Data

Three Questions, Three Routes

Questions

1. Are inventions spreading out?
2. Why are they spreading out?
3. What are the consequences?

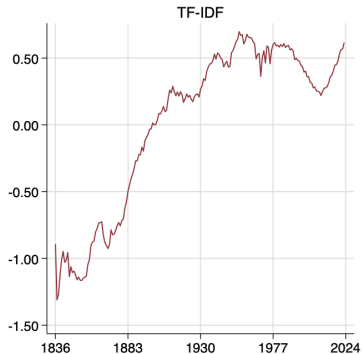
How we answer

1. Build and validate a **map of inventions**
2. Model how **entry and competition** determine spacing
3. Trace how spacing shapes **invention scale, spillovers, and growth**

Are Inventions Spreading Out?

Let's Check with Text

Standardized avg pairwise cosine similarity by year



Data: Claims text of 11M+ US patents

Standard Tool: **TF-IDF** word frequencies

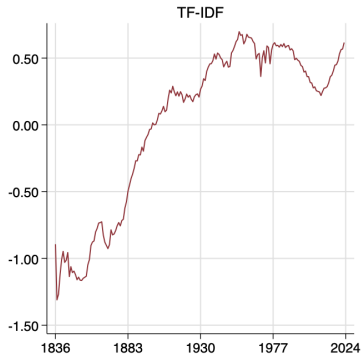
► Corpus

► Computation

► Standardization

Let's Check with Text

Standardized avg pairwise cosine similarity by year



Data: Claims text of 11M+ US patents

Standard Tool: **TF-IDF** word frequencies

Similarity is rising?

× Inventions appear to be *crowding*.

► Corpus

► Computation

► Standardization

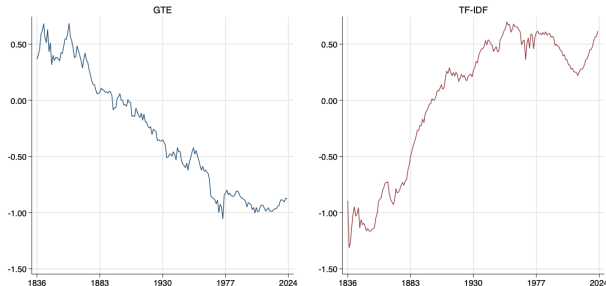
Same Text, Opposite Maps

Seven representations:

TF-IDF · GTE · PaECTER · S-BERT
Doc2Vec · USE · OpenAI

Divergent conclusions:

- TF-IDF: similarity *rising*
- Others: similarity *falling*



Which “map” should we trust?

Validation Across Judges, Scales, and Eras

Judge / benchmark	Scale	Era
Examiners / Interferences	Same invention	2001–2014
RAs / Pair comparisons	Relative similarity	1850–1975
Examiners / Classifications	Technology categories	1850–2023

► Interferences

► Human judgments

► Classifications

Validation Selects GTE Embeddings

		Same - invention collisions	Human rankings	Shared CPC categories	
		F10	Agreement	Section ROC	Class ROC
✓	GTE	0.90	0.62	0.596	0.656
	PaECTER	0.90	0.51	0.590	0.672
	S-BERT	0.82	0.54	0.600	0.671
×	TF-IDF	0.77	0.35	0.514	0.525

Higher is better in every column

✓ **Select GTE**

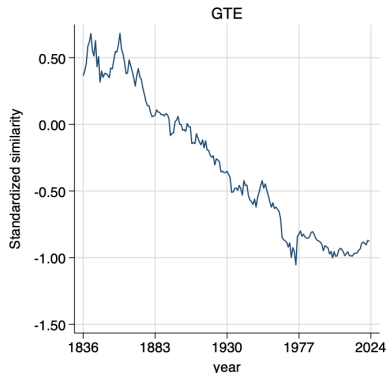
Consistent across benchmarks

× **TF-IDF ranks last**

On every benchmark

► bicycle ≠ velocipede

Validated Maps Agree: Inventions Are Spreading Out



About $1-1.5\sigma$ less similar

1838–2023, GTE

Three validated maps agree

GTE, PaECTER, S-BERT

Also locally, within fields, across cohorts & convex hull expanding

► Independent inventions

► Patent / entity counts

► Local / within fields

► Spatial scales

► Quantiles

► Within / between

► Class age

► Class / year FE

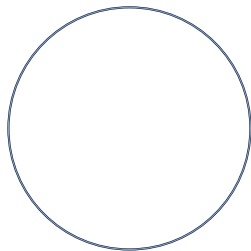
► Frontier

Why Are Inventions Spreading Out?

Spacing Gives Inventors Market Power

Idea space: circumference H

Similar problems, similar solutions



Spacing Gives Inventors Market Power

Idea space: circumference H

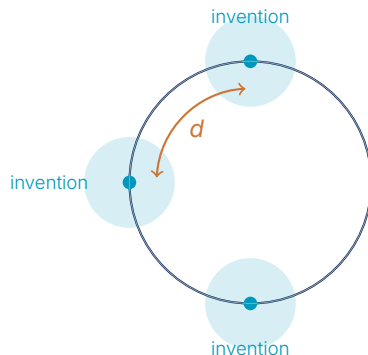
Similar problems, similar solutions

Inventors sell non-exclusive licenses

Pay entry cost f ; choose a location

Choose quality q_i and price p_i

$$\pi_i = R_i - c(q_i) - f$$



Spacing Gives Inventors Market Power

Idea space: circumference H

Similar problems, similar solutions

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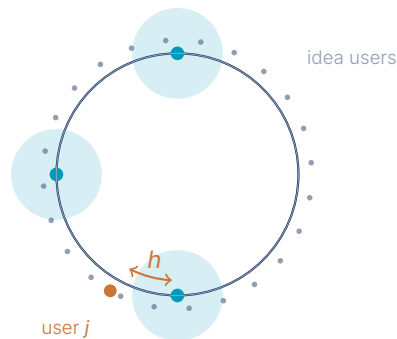
Choose quality q_i and price p_i

$$\pi_i = R_i - c(q_i) - f$$

Users buy one idea to improve TFP

Uniformly distributed on the circle

Net surplus: $Q_i - \tau h - p_i$ ▶ TFP



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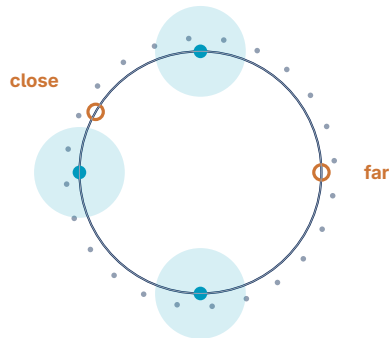
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Distance reduces close substitutes.



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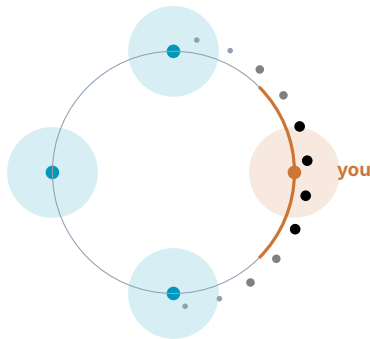
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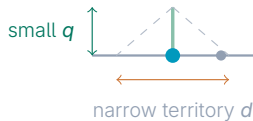
Net surplus: $Q_i - \tau h - p_i$ ► TFP

**Distance reduces close substitutes.
Isolation widens your market.**



A Larger Territory Requires a Larger Invention

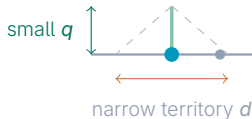
Close to neighboring inventor



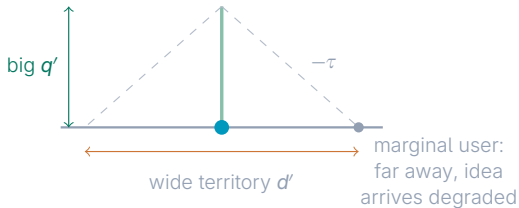
Your marginal user is $d/2$ away.

A Larger Territory Requires a Larger Invention

Close to neighboring inventor



Far from neighboring inventor



Your marginal user is $d/2$ away.

Wider spacing raises adaptation costs.

A larger quality gain makes the invention worth using farther away.

A wider territory calls for a larger invention.

Spacing Pins Prices and Quality

1. Price:

$$p^* = \tau d$$



► Detailed

► Existence

► $\tau\gamma > 1/2$

► Why spread?

► Comparative statics

► Equations

Spacing Pins Prices and Quality

1. Price:

$$p^* = \tau d$$

2. Quality investment:

$$q^* = d/\gamma$$

$$c(q) = \frac{1}{2}\gamma q^2$$

(fishing out)



► Detailed

► Existence

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Spacing Pins Prices and Quality

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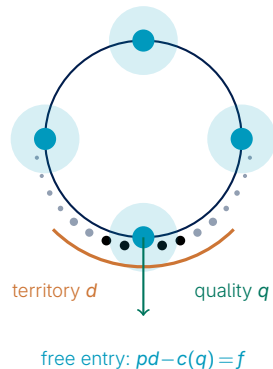
$$q^* = d/\gamma$$

$$c(q) = \frac{1}{2}\gamma q^2$$

(fishing out)

3. Free entry:

$$d^* = \sqrt{\frac{f}{\tau - 1/(2\gamma)}}$$



► Detailed

► Existence

► $\tau\gamma > 1/2$

► Why spread?

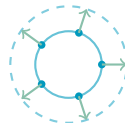
► Comparative statics

► Equations

Frontier Expansion Changes Equilibrium Spacing

Inventions open new directions Weitzman (1998)

$$\dot{H} = \delta n \left(Q - \frac{\tau d}{4} \right)$$



► Free entry

► Frontier law

► Scenarios

► Entry-cost curvature

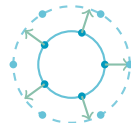
Frontier Expansion Changes Equilibrium Spacing

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Entry becomes more costly Jones (2009)

$$f(H) = \phi H^\alpha \quad \alpha = 1 \text{ in the baseline}$$



► Free entry

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Frontier Expansion Changes Equilibrium Spacing

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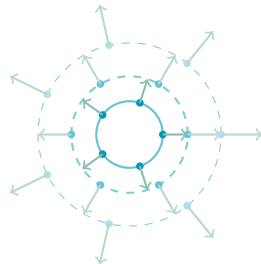
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$$f(H) = \phi H^\alpha \quad \alpha = 1 \text{ in the baseline}$$

Free entry requires wider spacing

$$d^* \uparrow \quad \text{and} \quad q^* = d^* / \gamma \uparrow$$



The Ripple: $H \uparrow \longrightarrow f \uparrow \longrightarrow d, q \uparrow \longrightarrow H \uparrow$

Inventors spread out and build bigger.

► Free entry

► Frontier law

► Scenarios

► Entry - cost curvature

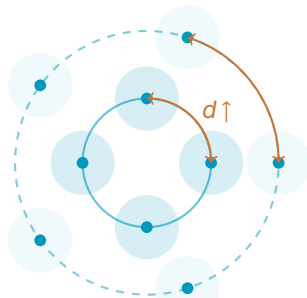
The Ripple's Fade: Attenuating Knowledge Spillovers

Quality gains include direct diffusion from local spillovers:

$$Q_i = \underbrace{q_i}_{\text{purchased idea}} + \underbrace{\beta\left(1 - \frac{d}{\lambda}\right)}_{\substack{\text{spillovers} \\ \text{(declines with } d\text{)}}} q$$

An endogenous brake on growth

Spreading weakens the contribution of neighboring inventions.



halos fade: $\beta(1 - d/\lambda) \rightarrow 0$

cf. Fort et al. 2026

Local TFP Gains, Economy - Wide R&D Costs

Nonrival ideas are **only locally useful** here (cf. Romer 1990).

R&D
productivity

$$\Pi \equiv \frac{g_{TFP}}{\text{Agg R\&D}}$$

cf. Bloom et al. 2020

Local TFP Gains, Economy - Wide R&D Costs

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R&D productivity

$$\Pi \equiv \frac{g_{TFP}}{\text{Agg R\&D}}$$

TFP: average local gains

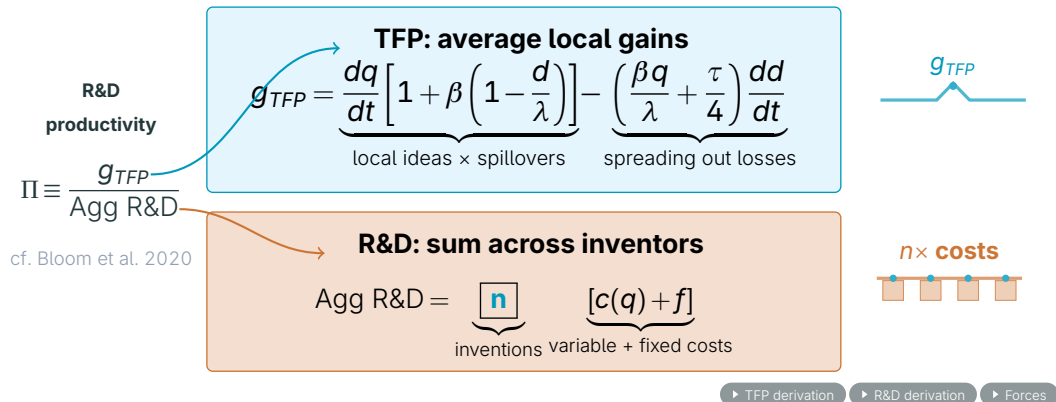
$$g_{TFP} = \underbrace{\frac{dq}{dt} \left[1 + \beta \left(1 - \frac{d}{\lambda} \right) \right]}_{\text{local ideas} \times \text{spillovers}} - \underbrace{\left(\frac{\beta q}{\lambda} + \frac{\tau}{4} \right) \frac{dd}{dt}}_{\text{spreading out losses}}$$

g_{TFP}

cf. Bloom et al. 2020

Local TFP Gains, Economy - Wide R&D Costs

Nonrival ideas are **only locally useful** here (cf. Romer 1990).



Equilibrium Spacing Enters the Growth Rate

Growth depends on the distance between inventions

$$g_H(t) = \underbrace{\delta \left(\frac{1}{\gamma} - \frac{\tau}{4} \right)}_{\text{R\&D and adaptation costs}} + \underbrace{\frac{\delta \beta}{\gamma} \left(1 - \frac{d(t)}{\lambda} \right)}_{\text{spillovers weaken as spacing rises}}$$

Wider spacing weakens spillovers and slows frontier growth.

Baseline frontier law; active spillovers ($d < \lambda$)

► Growth equations

► General model

► Equations

Growth depends on more than a scalar knowledge stock.

Equilibrium Positioning Shapes Growth

Growth depends on more than a scalar knowledge stock.

Positioning makes key growth objects equilibrium outcomes.

	Canonical treatment	Here: equilibrium spacing
Romer (1990)	Broad knowledge reuse	Local spillovers: $\beta(1 - d^*/\lambda)$
Jones (1995)	Research scale; imposed diminishing returns	Entry $n^* = H/d^*$; declining research productivity
Aghion–Howitt (1992)	Given innovation step size	Innovation scale: $q^* = d^*/\gamma$

What Are the Consequences?

One Geometry Unifies Many Facts

Spreading out

Patent similarity falls

This paper; Chiopris 2024

Spillovers weaken

Lucking+ 19; Fort+ 26;

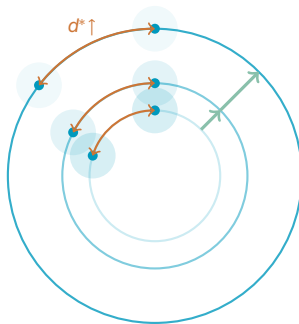
Berkes–Gaetani 26

Research productivity falls

Jones 95; Kortum 97; Bloom+ 20

Varieties increase

Patent counts; Hirschey+ 12



► Comparative statics

► Kelly replication

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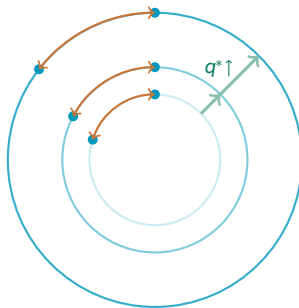
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Research productivity falls

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Building bigger

R&D rises

Hirschey+ 12

Breakthrough rates rise

Kelly+ 2021

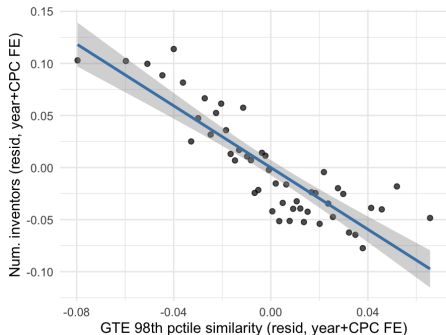
Patent values rise

Hall+ 05; Bessen+ 18; Kogan+ 17

► Comparative statics

► Kelly replication

The Model's Fingerprint: Spacing Couples with Quality



Co-inventors vs. similarity
residualized on year × CPC class

Wider spacing predicts larger inventions:

$$q^* = d/\gamma$$

More isolated inventions:

- Use **larger teams**
- More often **firm-assigned**
- More often **cited**
- Have **higher market value**

(KPSS: ≈\$22–33M per σ of isolation)

Within class–year; consistent with the mechanism.

► All four proxies

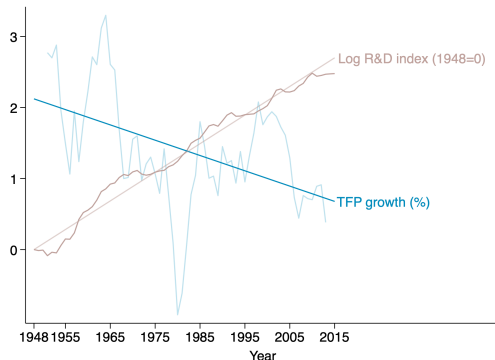
► Regression table

Research Productivity Falls Through Two Margins

$$g_{\Pi} = \underbrace{g_{g_{TFP}}}_{-1.6\%/yr} - \underbrace{g_{R\&D}}_{+4.0\%/yr} = \boxed{-5.6\%/yr}$$

How much of the **TFP-growth slowdown** comes from spreading?

How much of **R&D growth** comes from **novel** spatial forces?

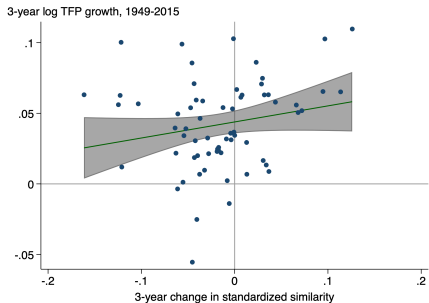


Bloom, Jones, Van Reenen and Webb, 2020

Accounting: Quantify the spatial contribution to each margin, then add them.

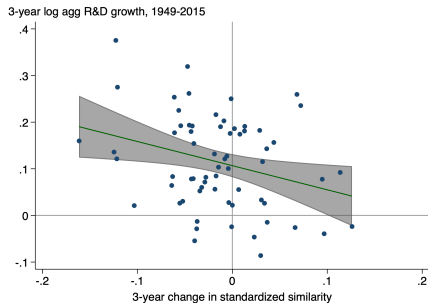
The Aggregate Patterns Match the Model's Predictions

TFP margin



More spreading, slower TFP growth

R&D margin



More spreading, faster R&D growth

These relationships also discipline the quantitative exercise.

Two Regressions Inform the Accounting

	What we estimate	Role in the accounting
TFP regression	TFP growth on spreading, spacing $\times$ spreading, and a time trend	Quantifies the TFP contribution directly
R&D regression	R&D growth on spreading and a time trend	Checks the variable - cost share and frontier growth used in calibration

Calibration $\longrightarrow$ Decompose aggregate R&D growth

entry expansion + quality scaling + fishing out + burden of knowledge

► TFP regression

► R&D regression

Separate Estimates Give Similar Magnitudes

External comparisons

TFP effect, 1981–2001

Regression

−0.16 pp/yr

Benchmark

−0.15 pp/yr

Bloom+2013 quasi-experiment

Variable-cost share

0.719

0.69

NSF survey

Internal consistency checks

Frontier growth

2.70%/yr

2.67%/yr

Regression / model identity

Spacing growth / R&D growth

0.2–0.4

1/3

Data / model

▸ TFP estimates

▸ TFP comparison

▸ R&D estimates

▸ Internal restrictions

From Calibration to R&D Components

Novel spatial channels

Entry across expanding idea space $(1 - \alpha/2)g_H$

Quality investment for wider territories $(\theta\alpha/2)g_H$

Familiar cost channels

Fishing out $(\theta\eta\alpha/2)g_H$

Burden of knowledge $(1 - \theta)\alpha g_H$

Cost share $\theta = \mathbf{0.69}$ (NSF) Frontier growth $g_H = \mathbf{2.67\%/yr}$ ($\frac{2}{3} \times 4.0\%$)

$\alpha = 1$: linear entry costs $\eta = 1$: quadratic R&D costs

► R&D growth equation

Baseline Model Accounting: A 42% Spatial Share

Component	pp/yr
TFP - growth deceleration	−1.60
Spatial contribution	−0.11
Other factors	−1.49
Aggregate R&D growth	+4.00
Entry across expanding idea space	+1.33
Quality investment for wider territories	+0.92
Fishing out	+0.92
Burden of knowledge	+0.83
R&D productivity decline	−5.60

$$\text{Spatial share: } \frac{0.11 + 1.33 + 0.92}{5.6} = 42\%$$

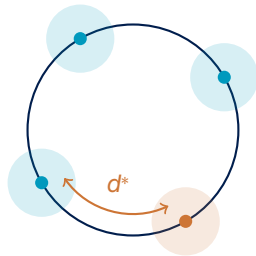
Sensitivity: up to 58%

The Spatial Equilibrium of Invention Matters for Growth

Inventors choose research directions in response to competitors.

Their spacing shapes market power, quality investment, and access to neighboring knowledge.

The resulting spatial equilibrium shapes growth.



Three Takeaways

1

Inventions spread out

Validated patent text and interference records.

2

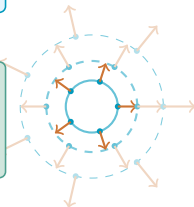
Inventors' equilibrium positioning shapes growth

Spacing shapes innovation scale and effective spillovers.

3

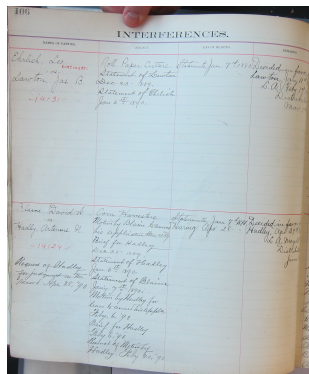
Spatial forces: 42% of the R&D productivity decline

Where inventors stand matters for growth



Backup Slides

Backup: Interference Data



INTERFERENCES.			
NAME OF PARTY	DATE	DATE OF HEARING	DATE OF DECISION
Ehrlich, Lee Lawton, Jas B - (1873) -	Roll Paper Cutters Statement of Lawton Dec. 22, 1890 Statement of Ehrlich Jan 6 & 7, 1891	Decided Jan 7 1890 Lawton, May 1891 Ehrlich, May 1891 Decided Jan 7 1890	
Blaine, David H Hadley, William R - (1874) -	Corn Harvesters Statement of Blaine Dec. 22, 1890 Brief for Hadley Dec. 22, 1890 Statement of Hadley Jan 6 & 7, 1891 Statement of Blaine Jan 7 & 8, 1891 Motion of Hadley for leave to amend his paper Feb. 6, 1891 Brief for Hadley Feb. 6, 1891 Rehearsal of Blaine Hadley Feb. 6, 1891	Decided Jan 7 1890 Hadley, May 1891 Blaine, May 1891 Decided Jan 7 1890	

Example (Jan 7, 1890)

Ehrlich v. Lawton: Roll paper cutters

Blaine v. Hadley: Corn harvesters

Purpose - digitized from National Archives:

- USPTO *Registers of Interferences*, 1864–1900
- 19,388 interference cases documented
- Average 504/year (2.7–4.9%)

Other sources:

- Case Files (1838–1900): 87–880/year
- DiSimone et al. (1950–1962): 640/year (1.4%)
- Calvert & Sofocleous (1980–1994): 237/yr (0.3%)
- Ganguli et al. (1998–2014): 76/year (0.05%)

Backup: Patent Corpus Details

Historical (1836–1975): ProQuest Patents Core

- OCR - digitized patent images
- Full text of claims extracted
- Quality varies with original document condition

Modern (1976–2023): USPTO PatentsView

- Machine - readable full text
- Structured data with claim parsing
- Consistent quality

Potential discontinuity at 1976:

- Some evidence of break in levels
- Trends consistent across periods
- Results robust to excluding transition years

Backup: Computing Similarity Efficiently

Challenge: $O(N^2)$ pairwise comparisons infeasible for millions of patents

Solution: For unit-normalized vectors, average cosine similarity reduces to:

$$\bar{S} = \frac{1}{N(N-1)} \sum_{i \neq j} \cos(v_i, v_j) = \frac{\|\sum_i v_i\|^2 - N}{N(N-1)}$$

Complexity: $O(N \cdot d)$ where d = embedding dimension

Implementation:

1. Normalize all vectors to unit length
2. Sum vectors: $S = \sum_i v_i$
3. Compute $\|S\|^2$
4. Apply formula

Cross-sectional SD: Subsample up to 10,000 patents/year

Backup: Alternative Normalizations

Main specification: Standardize by annual cross-sectional SD

Robustness checks:

1. Time-invariant global SD → Nearly identical results
2. Raw similarity (no standardization) → Same qualitative pattern
3. Different sample sizes per year → Robust

Why standardize?

- Different representations have different scales
- No intrinsic economic interpretation of raw similarity
- SD provides meaningful units for comparison

Backup: Interference Validation Task

Patent interferences (2001–2014):

- **First to invent:** USPTO proceeding for multiple applicants w/ identical claims
- Provides ground truth for “identical” similarity
- 322 true interfering pairs among 96,580 application pairs

Economic intuition: Examiner ranks pairs by similarity, investigates above threshold

- Higher threshold → fewer false positives but miss true interferences
- Lower threshold → catch more but burden staff with unnecessary investigations

Metrics:

- F10: Weights recall 10× more than precision (missing interferences is costly)
- PR AUC: Precision-Recall area under curve across all thresholds

Key result: GTE, PaECTER, OpenAI retrieve ~90% of true interferences with 2–5× fewer false positives than TF-IDF/S-BERT

Backup: Human Annotation Task

Historical patents (1850–1975):

- Sampled patent pairs that each model ranked at least 50 percentiles apart
- Annotators rank **relative** similarity of 2 patent pairs
- Tests temporal robustness (historical language): Oversample 1880–1920

Task: Do model rankings agree with human rankings?

- For each patent, rank others by similarity
- Compare model ranking to human ranking

Metric: Agreement coefficient from regression

$$\text{Human Rank} = \alpha + \beta \cdot \text{Model Rank} + \epsilon$$

Higher β = better agreement

Backup: Classification Validation Task

USPTO Classifications (1850–2023):

- CPC technology codes assigned by examiners
- Section level (8 categories) and Class level (120+ categories)
- Captures expert judgment of technological relatedness

Task: Predict whether patent pair shares classification

- Same Section (coarse): 8 top-level categories
- Same Class (fine): 3-digit classification

Metric: ROC AUC

- Area under Receiver Operating Characteristic curve
- 0.5 = random, 1.0 = perfect

Backup: Why TF-IDF Fails

TF-IDF overweights period-specific language:

- Treats "velocipede" (1880s) and "bicycle" (modern) as unrelated
- Period-specific terminology dominates similarity scores
- Creates spurious correlation with time

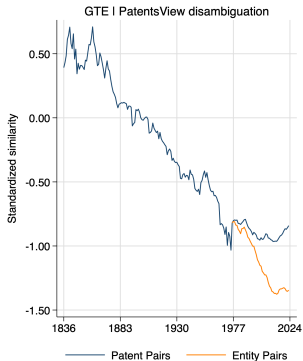
Example: 1880 velocipede patent

- TF-IDF: High similarity to other 1880s patents (shared vocabulary)
- GTE: High similarity to modern bicycle patents (shared concepts)

Evidence:

- TF-IDF similarity correlates with word overlap
- GTE similarity correlates with conceptual similarity
- Google Ngrams shows vocabulary shifts over time

Backup: Robustness — Multi-Patent Entities



Concern: Post-2000 dynamics coincide with: business method patents, non-practicing entities, increased defensive patenting.

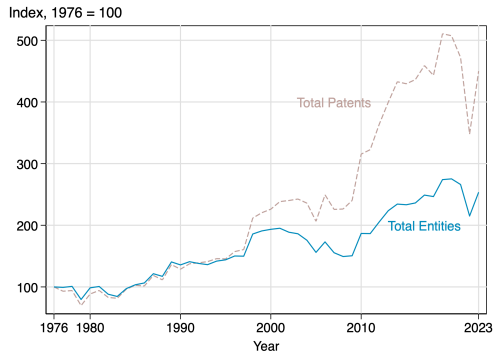
- **Multiple patents from same entity may be similar but not independent.**

Strategy: Sample 1 patent/entity-year

Result:

- Decline persists after correction
- **Independent inventions** still spreading out

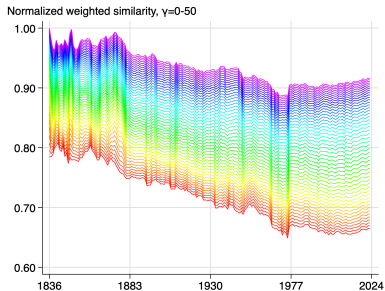
Backup: Growth in Patents vs. Patenting Entities



- Number of issued utility patents and unique patenting entities per year
- Divergence after 1999: substantial growth in patents per entity
- Coincides with business method patents and non-practicing entities
- Motivates sampling 1 patent per entity per year for robustness

Backup: Spreading Occurs Nearby and Within Fields

Even among nearby inventions



Greater weight on near neighbors

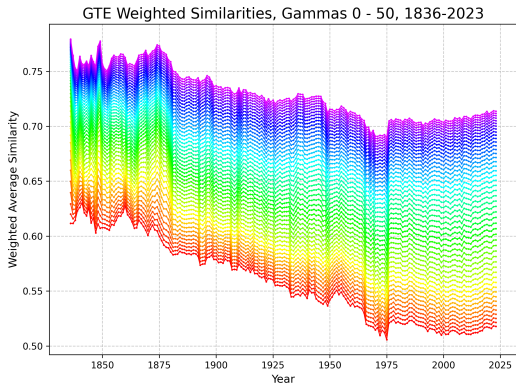
Even within technology classes



Within three-character CPC classes

The model needs to explain local spacing

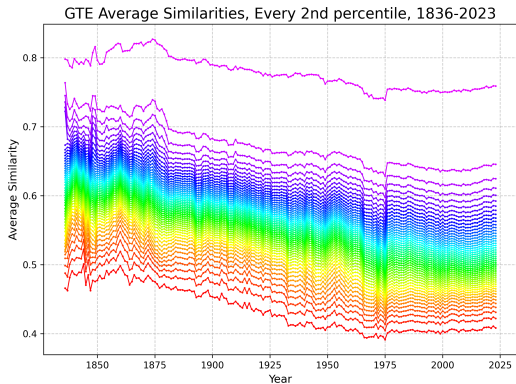
Backup: Similarity at Different Spatial Scales



Weighted average: $\equiv \frac{1}{n} \sum_{i=1}^n \frac{\sum_{j \neq i} (1-d_{ij}) e^{-\gamma d_{ij}}}{\sum_{j \neq i} e^{-\gamma d_{ij}}}$
where γ from 0 (global) to 50 (local)

- **Key finding:** Similar declining trends across all spatial scales
- Model predictions concern averages — important to verify pattern holds across distribution
- Post-2000 arrest slightly stronger at local scales (consistent with entity correction)

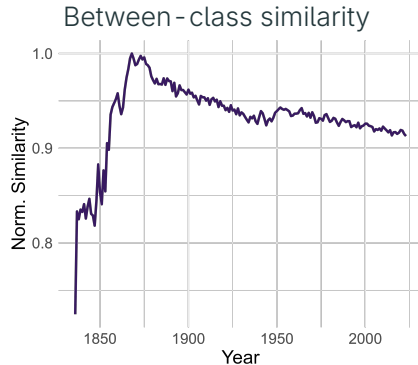
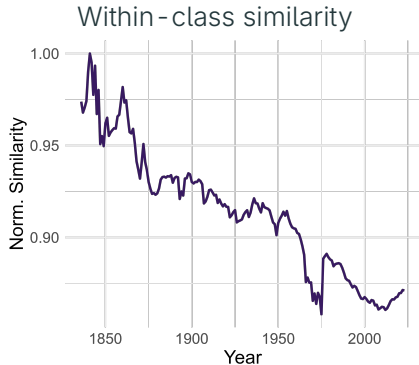
Backup: Similarity at Different Quantiles



Similarity at Different Quantiles:

- 50 quantiles of pairwise similarity in each year
- Secular decline is robust across all quantiles
- Post-2000 increase in similarity is slightly faster for higher quantiles

Backup: Within vs. Between Technology Classes



- **Addresses compositional concern:** Decline not driven by shifts across technology fields — spreading out occurs *within* established classes

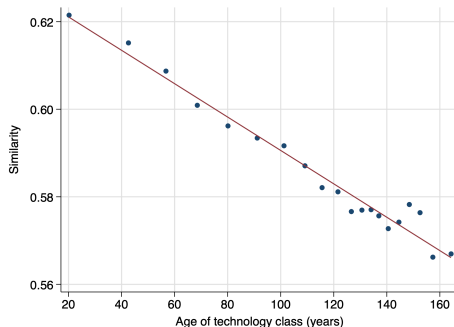
Backup: Robustness — Spreading Out Within Technology Classes

Alternative explanations: Changing patent office practice over time? Shifts across major technology areas?

Test: Within-class similarity by class “age”

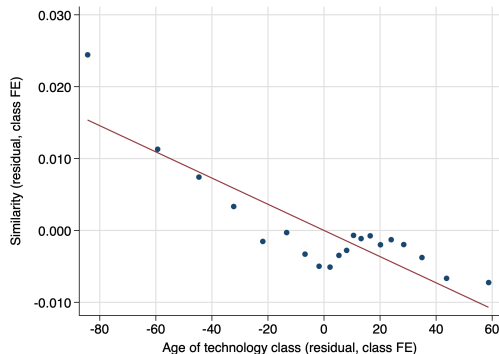
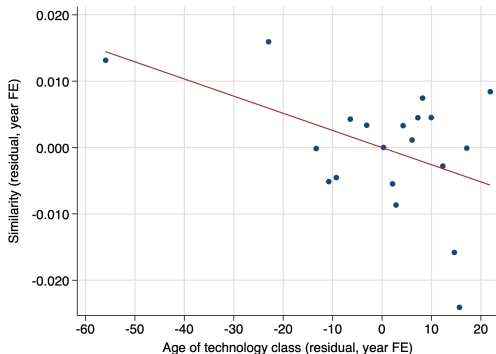
- Birth = Class first issued 50 patents
- e.g., Combinatorial Chemistry 2001
- Addresses compositional concerns

Finding: Within-class similarity declines as classes mature



Spreading out is a dynamic process tied to field evolution

Backup: Similarity Within Class by Class Age

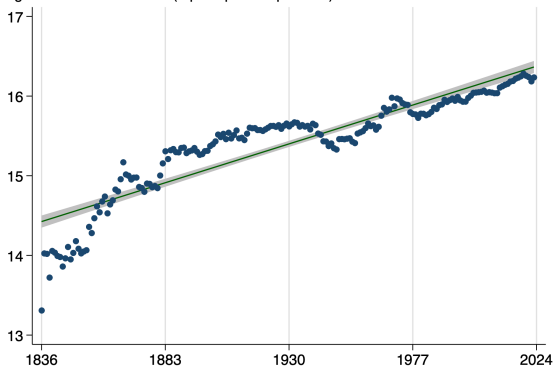


Left: Cross-cohort comparison (year FE) — robustness to patent office practice

Right: Within-cohort comparison (class FE) — robustness to composition

Corroboration: Expanding Convex Hull

Log volume of convex hull (7 principal components)



Log TFP Specification

Why linear in log TFP? $A_i(h) = Q_i - \tau h$

Standard in spatial competition (Salop 1979):

- Idea consumers have preferences linear in quality net of distance costs
- $A_i(h)$ interpreted as log TFP $\Rightarrow$ firms care about *proportional* productivity gains

Microfoundation: Each downstream firm has one unit of fixed input ℓ and produces:

$$y = e^A \cdot \ell$$

With output price = 1 and $\ell = 1$, profit is $\pi = e^A$. Willingness to pay for technology delivering incremental log TFP A (relative to baseline $e^0 = 1$):

$$WTP = e^A - 1 \approx A \quad (\text{first-order Taylor approximation})$$

Accuracy: For annual TFP increments ($A \approx 0.015/\text{year}$), approximation error $< 0.01\%$

Advantage: Predictions directly comparable to empirical TFP elasticities (Bloom et al. 2013) and growth accounting (Bloom et al. 2020)

Equilibrium: Pricing and Quality

Symmetric equilibrium: n inventors, equal spacing $d = H/n$, identical (p, q)

1. Pricing. Boundary firm at $\tilde{h}$ indifferent between inventor i and neighbor:

$$Q - p_i - \tau \tilde{h} = Q - p - \tau(d - \tilde{h}) \Rightarrow \tilde{h} = \frac{d}{2} + \frac{p - p_i}{2\tau}$$

Revenue $R_i = 2p_i \tilde{h}$; FOC $\partial R_i / \partial p_i = 0$ yields: $p^* = \tau d$

2. Quality. MR = MC for quality investment ($\partial Q_i / \partial q_i = 1$, $\partial \tilde{h} / \partial Q_i = 1/2\tau$):

$$\frac{p_i}{\tau} = \gamma q_i \Rightarrow q^* = \frac{d}{\gamma} \quad (\eta = 1)$$

3. Zero-profit condition. Substitute $R = \tau d^2$, $c(q^*) = \frac{d^2}{2\gamma}$:

$$\tau d^2 - \frac{d^2}{2\gamma} - f = 0 \Rightarrow d^* = \sqrt{\frac{f}{\tau - \frac{1}{2\gamma}}}$$

Backup: Equilibrium Existence Conditions

Spreading - out condition:

$$\tau\gamma > \frac{1}{2}$$

Marginal revenue of expanding territory exceeds marginal cost.

Second-order conditions:

- Pricing: $\partial^2 R / \partial p^2 < 0$ (satisfied)
- Quality: $\partial^2 \pi / \partial q^2 = -\gamma < 0$ (satisfied)
- No spatial deviation (verified in paper)

Additional conditions:

- Spillover reach: $d < \lambda$ (spillovers active)
- Full coverage: All downstream firms adopt some technology

Backup: Spreading Out — Proposition

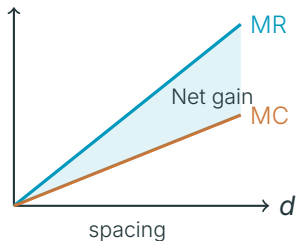
Proposition (Spreading Out)

For $\tau\gamma > \frac{1}{2}$, equilibrium spacing increases with opportunity space: $\frac{dd^*}{dH} > 0$.
Inventions become **less similar** over time.

Adaptation costs must create sufficient pricing power

Backup: Spreading Out — Why Is It Profitable?

Why is spreading out profitable?



Marginal revenue of expanding territory:

- Revenue $R = \tau d^2 \Rightarrow MR = 2\tau d$

Marginal cost of expanding territory:

- Need higher quality: $q = d/\gamma$
- $MC = d/\gamma$

Spreading profitable when:

$$MR > MC \Rightarrow \boxed{\tau\gamma > \frac{1}{2}}$$

Adaptation costs must create sufficient pricing power

Backup: Comparative Statics Derivations

Spreading out: From zero-profit condition $d^2(\tau - \frac{1}{2\gamma}) = \phi H$:

$$\frac{dd}{dH} = \frac{\phi}{2d(\tau - \frac{1}{2\gamma})} = \frac{\phi}{dR/dd - dc/dd} > 0$$

Rising quality and prices:

$$\frac{dq}{dH} = \frac{1}{\gamma} \frac{dd}{dH} > 0, \quad \frac{dp}{dH} = \tau \frac{dd}{dH} > 0$$

Rising entry:

$$\frac{dn}{dH} = \frac{1}{d} - \frac{H}{d^2} \frac{dd}{dH} > 0 \text{ under spreading-out condition}$$

Declining R&D productivity: Local TFP gains contrast with economy-wide R&D costs.

Backup: Model Equations

Fixed cost (burden of knowledge):	$f(H) = \phi H^\alpha$
R&D cost:	$c(q_i) = \frac{1}{2}\gamma q_i^{1+\eta}$
Realized quality (with spillovers):	$Q_i = q_i + \frac{1}{2}\beta\left(1 - \frac{d}{\lambda}\right)(q_{i-1} + q_{i+1})$

Equilibrium pricing and quality:	$p^* = \tau d, \quad q^* = \frac{d}{\gamma}$
Equilibrium spacing:	$d^*(H) = \sqrt{\frac{\phi H}{\tau - \frac{1}{2\gamma}}}$
Equilibrium entry:	$n^* = \frac{H}{d^*}$
Equilibrium revenue:	$R^* = p^* \cdot d^* = \tau d^2$

Growth rates (baseline $\alpha = 1, \eta = 1$):	
Spacing, quality, entry:	$g_d = g_q = g_n = \frac{1}{2}g_H$
Fixed cost:	$g_f = g_H$
Aggregate R&D:	$g_{R\&D} = \frac{3}{2}g_H$

Backup: Rising Entry Costs Require Wider Territories

Revenue

$$R = pd = \tau d^2$$

Variable cost

$$c(q) = \frac{1}{2}\gamma q^2 = \frac{d^2}{2\gamma}$$

$\tau\gamma > 1/2$; both margins rise when $0 < \alpha < 2$

Backup: Rising Entry Costs Require Wider Territories

Revenue

$$R = pd = \tau d^2$$

Variable cost

$$c(q) = \frac{1}{2}\gamma q^2 = \frac{d^2}{2\gamma}$$

Free entry: revenue covers variable and fixed costs

$$d^2\left(\tau - \frac{1}{2\gamma}\right) = \phi H^\alpha$$

$\tau\gamma > 1/2$; both margins rise when $0 < \alpha < 2$

Backup: Rising Entry Costs Require Wider Territories

Revenue

$$R = pd = \tau d^2$$

Variable cost

$$c(q) = \frac{1}{2}\gamma q^2 = \frac{d^2}{2\gamma}$$

Free entry: revenue covers variable and fixed costs

$$d^2 \left(\tau - \frac{1}{2\gamma} \right) = \phi H^\alpha$$

Wider spacing

$$d \propto H^{\alpha/2}$$

More inventions

$$n = H/d \propto H^{1-\alpha/2}$$

$\tau\gamma > 1/2$; both margins rise when $0 < \alpha < 2$

Backup: Robustness to Entry Cost Curvature

With $f(H) = \phi H^\alpha$, equilibrium spacing satisfies $d \propto H^{\alpha/2}$:

Condition	Entry Growth	Prediction
$0 < \alpha < 2$	$g_n = (1 - \frac{\alpha}{2})g_H > 0$	Entry grows ✓
$\alpha = 2$	$g_n = 0$	Entry stagnates
$\alpha > 2$	$g_n < 0$	Entry declines ×

Main results robust for $\alpha < 2$:

- Spreading out: $g_d = \frac{\alpha}{2}g_H > 0$ for any $\alpha > 0$
- Declining R&D productivity: Holds throughout range
- Higher $\alpha \rightarrow$ faster spreading, but lower spatial share of productivity decline

Counterfactual boundary: Patent counts grow over time, ruling out $\alpha \geq 2$

Backup: A Knowledge Production Function

Evidence of expansion:

- Patent filings: $\approx 500/\text{yr}$ (1840s) $\rightarrow 350,000/\text{yr}$ (2020s)
- Each discovery opens **adjacent research directions** Weitzman 1998; Scotchmer 1991
- Electricity $\rightarrow$ electronics $\rightarrow$ computing $\rightarrow$ AI...

Formalizing the intuition:

$$\dot{H} = \delta n \underbrace{(Q - \tau d/4)^\nu}_{\text{usable quality opens new directions}}$$

- $\nu = 1$ baseline; $0 \leq \nu \leq 1$ governs quality-to-frontier conversion
- Q : gain above inherited knowledge (the paper's $Q - Q_0$)

Key features:

- Uses *social* quality Q (incl. spillovers), not private effort q
- Net of adaptation costs—only “usable” quality opens new directions
- Standard supply-side KPF (Romer 1990; Jones 1995; Kortum 1997)

Backup: How Entry Costs Shape Equilibrium

As idea space H grows, what happens to d^*, n^*, q^*, p^* ?

Scenario	d^*, q^*, p^*	Varieties n^*
1. f constant	unchanged	$\uparrow$ (linear in H)
2. $f(H)$ decreasing	$\downarrow$ (clustering!)	$\uparrow\uparrow$ (faster)
3. $f(H)$ increasing (KPF baseline)	$\uparrow$ (spreading)	$\uparrow$ (grows with H)
4. $f(H)$ increasing rapidly	$\uparrow\uparrow$ rapidly	$\downarrow$ (fewer)

Only rising H, f is consistent with spreading out and expanding varieties

Backup: Forces Reducing Research Productivity

Forces reducing g_{TFP} :

$$g_{TFP} = \frac{dA}{dH} \cdot \dot{H} = \left(\underbrace{\frac{dq}{dH} \left[1 + \beta \left(1 - \frac{d}{\lambda} \right) \right]}_{\text{(1) Spillover attenuation: mult.} \downarrow} - \underbrace{\frac{\beta q}{\lambda} \frac{dd}{dH}}_{\text{(2) Spillover base: } q \uparrow} - \underbrace{\frac{\tau}{4} \frac{dd}{dH}}_{\text{Adapt. drag}} \right) \cdot \underbrace{\left[g_H^* + \frac{\delta \beta}{\gamma} \left(1 - \frac{d}{\lambda} \right) \right]}_{\text{(3) Frontier spillovers} \downarrow \text{ as } d \uparrow} \cdot H$$

Forces increasing Agg R&D:

$$\frac{d(\text{Agg R\&D})}{dH} = \underbrace{\frac{dn}{dH} \cdot [c(q) + f(H)]}_{\text{(6) Entry expansion}} + \underbrace{n \cdot c'(q)}_{\text{(4) Fishing out}} \cdot \underbrace{\frac{dq}{dH}}_{\text{(7) Quality scaling}} + \underbrace{n \cdot f'(H)}_{\text{(5) Burden of knowledge}}$$

Spatial Forces: 1, 2, 3, 6, 7 — **Familiar cost channels: 4, 5**

Backup: Why the TFP Regression Has an Interaction

Quality gains raise TFP; weaker spillovers and adaptation losses reduce it.

$$g_{TFP} = \frac{dq}{dt} \left[1 + \beta \left(1 - \frac{d}{\lambda} \right) \right] - \left(\frac{\beta q}{\lambda} + \frac{\tau}{4} \right) \frac{dd}{dt}$$

Substitute equilibrium quality $q = d/\gamma$ and approximate $\dot{d}$ by Δd :

$$g_{TFP} = \left[\frac{1 + \beta}{\gamma} - \frac{\tau}{4} - \frac{2\beta}{\gamma\lambda} d \right] \Delta d$$

The marginal TFP gain from further spreading falls with existing spacing.

The coefficient on $d\Delta d$ is $-2\beta/(\gamma\lambda) < 0$.

Backup: From R&D Costs to the Regression

Aggregate spending sums variable and fixed costs across inventors.

$$R\&D = n[c(q) + f], \quad \theta = \frac{c(q)}{c(q) + f}$$

$$g_{R\&D} = g_n + \theta(1 + \eta)g_q + (1 - \theta)g_f$$

Equilibrium links these margins to frontier growth and spreading:

$$g_n = g_H - g_d, \quad g_q = g_d, \quad g_f = \alpha g_H$$

$$g_{R\&D} = \underbrace{[1 + \alpha(1 - \theta)]g_H}_{\text{frontier expansion}} + \underbrace{[\theta(1 + \eta) - 1]g_d}_{\text{quality costs less entry}}$$

Spreading raises R&D when $\theta(1 + \eta) > 1$.

Backup: TFP and R&D Growth Equations

TFP growth:

$$g_{TFP} = \underbrace{\dot{q} \left(1 + \beta - \frac{\beta d}{\lambda} \right)}_{\text{quality with spillovers}} - \underbrace{\left(\frac{\beta q}{\lambda} + \frac{\tau}{4} \right) \dot{d}}_{\text{spillover attenuation and adaptation}}$$

R&D growth:

$$g_{R\&D} = \underbrace{g_n}_{\text{Entry}} + \underbrace{\theta \cdot g_q}_{\text{Quality scaling}} + \underbrace{\theta \cdot g_q}_{\text{Fishing out}} + \underbrace{(1-\theta)g_f}_{\text{Burden of knowledge}}$$

where θ = variable cost share; $\alpha = 1$, $\eta = 1$

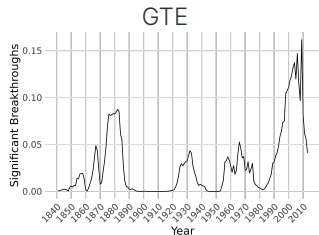
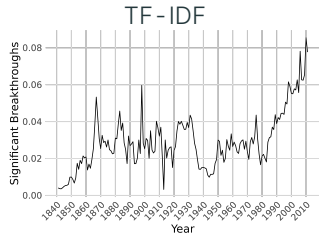
Backup: Growth Equations with General Cost Curvatures

Component	General	Baseline ($\alpha = 1, \eta = 1$)
Entry cost	$f(H) = \phi H^\alpha$	ϕH
R&D cost	$c(q) = \frac{1}{2}\gamma q^{1+\eta}$	$\frac{1}{2}\gamma q^2$
Spacing growth	$g_d = \frac{\alpha}{2}g_H$	$\frac{1}{2}g_H$
Quality growth	$g_q = g_d$	$\frac{1}{2}g_H$
Entry growth	$g_n = (1 - \frac{\alpha}{2})g_H$	$\frac{1}{2}g_H$
Fixed cost growth	$g_f = \alpha g_H$	g_H

R&D growth equation:

$$g_{R\&D} = \underbrace{(1 - \frac{\alpha}{2})g_H}_{\text{Entry}} + \underbrace{\frac{\theta\alpha}{2}g_H}_{\text{Quality scaling}} + \underbrace{\frac{\theta\eta\alpha}{2}g_H}_{\text{Fishing out}} + \underbrace{(1 - \theta)\alpha g_H}_{\text{Burden of knowledge}}$$

Backup: Kelly et al. Breakthrough Replication



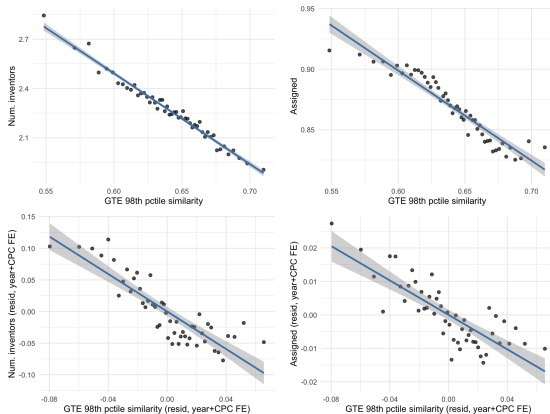
Kelly et al. (2021): Identify “breakthrough” patents using similarity to future patents

Our replication with GTE:

- Qualitative conclusions align (more breakthroughs today)
- Quantitative results more robust (less sensitivity to methodological choices)
- TF-IDF produces noisier breakthrough classification

Implication: Validated similarity measures improve downstream analyses

Backup: Spacing Predicts Investment and Quality



Model: $q^* = d/\gamma$ — more isolated inventors invest more in quality

4 q proxies: Co-inventors, assigned patents, 5-yr citations, KPSS patent value

Confirmed in cross-section and within technology classes over time

Spacing and Quality: Regression Results

	Co-Inventors	Firm Assignment	Citations (5-yr)	KPSS Value
<i>Panel A: Cross-sectional, mean similarity</i>				
GTE	−2.545*** (0.679)	−0.659*** (0.180)	6.347 (4.480)	−22.093** (9.036)
PaECTER	−4.928*** (1.002)	−1.723*** (0.424)	−36.681*** (8.383)	−122.904*** (46.101)
<i>Panel B: Cross-sectional, 98th percentile similarity</i>				
GTE	−1.477** (0.670)	−0.253* (0.131)	4.788 (3.220)	−32.531*** (7.022)
PaECTER	−1.067 (1.592)	−1.174*** (0.431)	−19.404*** (6.443)	−84.301* (43.556)
Observations	219,772	219,772	219,772	3,274,108
Fixed Effects	Year + CPC Class			
<i>Panel C: Time-series, Δ mean similarity, CUSP (1836–2023)</i>				
GTE	−0.011* (0.005)	—	—	—
PaECTER	−0.004** (0.001)	—	—	—
Observations	15,813			
Fixed Effects	Year + CPC Class			

Backup: TFP Regression

Model: use $q = d/\gamma$ to eliminate unobserved quality.

$$g_{TFP} = \left[\frac{1+\beta}{\gamma} - \frac{\tau}{4} - \frac{2\beta d}{\gamma\lambda} \right] \Delta d$$

Regression: the effect of spreading depends on existing spacing.

$$\Delta_h \log TFP_t = b_0 + b_1 \Delta_h D_t + b_2 D_{t-1} \Delta_h D_t + b_3 t + \epsilon_t$$

Interpretation: $b_1 + b_2 D$ is the marginal effect of spreading.

Annualized spatial contribution: $(\hat{b}_1 + \hat{b}_2 D) \overline{\Delta_h D} / h$.

Backup: R&D Regression

Model: frontier expansion and spreading determine spending growth.

$$g_{R\&D} = [1 + \alpha(1 - \theta)]g_H + [\theta(1 + \eta) - 1]g_d$$

D_t : negative standardized similarity; t : years from 1991. The trend proxies frontier growth.

$$\Delta_h \log(R\&D)_t = a_0 + a_1 t + a_2 \Delta_h D_t + \epsilon_t$$

Slope: variable - cost share

$$\hat{\theta} = \frac{\hat{a}_2 + 1}{1 + \eta} = 0.719$$

Intercept: frontier growth

$$\hat{g}_H^{1991} = \frac{\hat{a}_0/h}{1 + \alpha(1 - \hat{\theta})} = 2.70\%/yr$$

Backup: TFP Regression Results

	Annual		3 - Year	5 - Year
$b_1: -\Delta\text{Sim}$	-0.169*** (0.057)	-0.171** (0.083)	-0.278*** (0.095)	-0.269*** (0.098)
$b_2: \text{interaction}$	—	-0.015 (0.342)	-0.408 (0.320)	-0.571* (0.312)
$b_3: \text{Year}$	-2.67e-4*** (9.90e-5)	-2.62e-4** (9.99e-5)	-8.19e-4*** (1.91e-4)	-1.19e-3*** (2.70e-4)
<i>Annualized TFP drag, pp/year</i>				
1948	—	-0.082	-0.068	-0.041
1991	-0.084	-0.085	-0.139	-0.156
2015	—	-0.084	-0.127	-0.136

Standard errors in parentheses. Constant included. TFP: Bloom et al. (2020), 1948–2015.

Backup: Validation of Spillover Elasticity

Quasi-experimental estimates of R&D spillovers Bloom et al. 2013, Lucking et al. 2019

- Firm-level TFP elasticity to shocks to spillover pool
- SPILLTECH $\equiv$ sum of neighbors' R&D, weighted by idea distance
- IV: State R&D tax credit shocks

Ingredients:

- Bloom et al. 2013 (1981-2001): $\hat{\beta} = \mathbf{0.206}$
- Standard deviation $\log(\text{SPILLTECH}) = \mathbf{1.04}$
- Avg change in similarity (Bloom sample period): $\mathbf{-0.007\sigma/\text{yr}}$

Implied TFP drag from average rate of spreading out:

- $-0.007 \times 1.04 \times 0.206 = \mathbf{-0.15\%/yr}$ ✓

Backup: R&D Regression and Implied Parameters

	Annual	3 - Year	5 - Year
α_2 : $-\Delta\text{Sim}$	0.165 (0.177)	0.448** (0.219)	0.438* (0.244)
α_1 : Year (1991 = 0)	$-2.88\text{e-}4$ ($3.05\text{e-}4$)	$-7.79\text{e-}4$ ($6.58\text{e-}4$)	$-1.65\text{e-}3^*$ ($9.65\text{e-}4$)
α_0 : Constant	0.034*** (0.006)	0.102*** (0.013)	0.173*** (0.018)
Implied variable-cost share θ	0.583	0.724	0.719
Implied frontier growth g_H^{1991}	2.40%/yr	2.66%/yr	2.70%/yr

Parameters use annualized intercepts and the approximate similarity-spacing mapping.

Standard errors in parentheses. R&D: Bloom et al. (2020), 1948–2015.

Backup: Internal Restrictions on Frontier and R&D Growth

Baseline restrictions: $\alpha = \eta = 1$, $g_d = \frac{1}{2}g_H$.

$$g_{R\&D} = (2 - \theta)g_H + (2\theta - 1)g_d = \frac{3}{2}g_H$$

Two routes to frontier growth

$$g_H = \frac{2}{3} \times 4.0\% = 2.67\%/yr$$

$$\hat{g}_H^{1991} = \frac{0.173/5}{2 - 0.719} = 2.70\%/yr$$

Spacing relative to R&D growth

$$\frac{g_d}{g_{R\&D}} = \frac{g_H/2}{3g_H/2} = \frac{1}{3}$$

Empirical proxy ratio: **0.2–0.4**

Backup: Entry Costs Govern the Spatial Share

Dominant uncertainty: α (entry cost curvature)

Entity growth suggests $\alpha \approx 0.67$ unique inventor growth
(2.0%/yr)

$\eta = 0.625$ from quasi-experiment Guceri-Liu 2019

Full parameter range: 33–58%

42–58% when $\alpha \leq 1$.

Entity growth proxies invention growth.

α	Spatial share (%)	
	$\eta = 1$	$\eta = 0.625$
0.50	55	58
0.75	48	51
1.00	42	46
1.25	37	41
1.50	33	37

Backup: From TFP Drag to the Accounting Contribution

The five-year regression implies a worsening spatial drag

$$\underbrace{-0.041}_{1948} \rightarrow \underbrace{-0.136}_{2015} \quad \text{pp/year}$$

The change accounts for 6.8% of the TFP-growth slowdown

$$\frac{0.136 - 0.041}{1.4} = 6.8\%$$

Contribution to the decline in research productivity

$$6.8\% \times (-1.6\%/yr) = -0.11\%/yr$$